



THE UNIVERSITY OF
SYDNEY

DATA1002 Stage 1 Report

Tutorial Section: **36**

Group Number: **8**

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February 9, 2026

1 Define Topic

This study investigates how variations in Australia’s energy system and climatic conditions collectively affect CO₂ emission levels between 2010 and 2022. Our analysis focuses on three dimensions:

- Petroleum consumption as an indicator of energy demand in transportation and industry;
- The growth of renewable generation through hydropower, wind, and solar energy;
- Climatic factors such as rainfall and evapotranspiration that regulate carbon fluxes in agricultural and soil systems.

Australia plays a disproportionate role in global emissions. The country has a small population but contributes heavily to fossil-fuel-related CO₂, much of it through exports. Its domestic energy system also remains reliant on fossil sources (Grant & Hare, 2024; Li et al., 2020). This reliance increases policy pressure to meet climate targets and exposes energy-intensive sectors to transition risks. Yet, Australia’s large renewable potential offers major opportunities for diversification. Hydropower, wind, solar, and waste-to-energy are promising but remain underused (Li et al., 2020).

The significance of this study lies in showing how primary energy demand, renewable growth, and hydrological factors have interacted to shape CO₂ emissions over the past decade. Such evidence is necessary for policy makers aiming to design effective decarbonization pathways, especially since projected fossil exports could use up a large share of the global carbon budget (Grant & Hare, 2024). For industry, recognising the volatility of fossil demand and the stabilising effect of renewables and evapotranspiration supports better risk management and long-term investment planning (Metz et al., 2023).

2 Data Description

2.1 Cape Grim Greenhouse Gas Data (CO₂)

Data Provenance

This dataset was obtained from the Commonwealth Scientific and Industrial Research Organisation (CSIRO) and the Australian Bureau of Meteorology via the Kennaook / Cape Grim Baseline Air Pollution Station on 30 August 2025 (CSIRO & Bureau of Meteorology, 2025). Spanning the period from May 1975 to July 2025, the dataset provides monthly mean concentrations of the greenhouse gas carbon dioxide (CO₂), which are derived through analysis of the composition of air masses arriving at Kennaook / Cape Grim after long trajectories over the Southern Ocean.

Data Dictionary

The dataset comprises 591 entries across eight columns. Table 1 presents an overview of the attributes of the dataset, including their data types and corresponding descriptions.

Attribute	Data type	Description
YYYY	Integer	Year of measurement
MM	Integer	Month of measurement
DD	Integer	Day of measurement
DATE	Float	Decimal year
CO ₂ (ppm)	Float	Monthly mean atmospheric CO ₂ concentration
GR (ppm / year)	Float	Annual growth rate of atmospheric CO ₂
Source	Object	Measurement method, with <i>in situ</i> indicating continuous real-time monitoring and <i>flask</i> indicating laboratory analysis of flask air samples

Table 1: Data dictionary for Cape Grim Greenhouse Gas Data (CO₂).

Limitations

Missing values in the growth rate (GR) variable were handled using mean imputation, which may smooth over short-term variability. In addition, differences in measurement methods (*in situ* vs. *flask*) in CO₂ growth rate may introduce comparability issues, such as differences in precision and calibration.

2.2 Rainfall and Evaporation

Data Provenance

Weather situation dataset, obtained from the Queensland government’s open data platform (Queensland Government, 2025) on September 5th, contains 9,381 daily records from Burinjuck Dam (2000–2025). It includes station, rainfall, evaporation, and other variables. This part focuses on rainfall and evapotranspiration as indicators of crop growth and CO₂ absorption and release.

Data Dictionary

Attribute	Data type	Description
date	String	Daily date (YYYY-MM-DD)
rainfall_mm	Float	Daily rainfall (mm)
evaporation_mm	Float	Actual evapotranspiration estimated by Morton’s method (mm)

Table 2: Data dictionary for Rainfall and Evaporation Data.

Limitations

This dataset only reflects the local climate conditions of specific regions in Australia, which may vary greatly from the national climate. And as shown, only three variables will be explored (table 2), this may lead to a selection bias. Daily data can capture details of extreme events that may be lost when aggregated monthly. Furthermore, climate factors do not directly produce CO₂, but they affect emissions through energy use and vegetation growth, which may introduce false correlations.

2.3 IEA Monthly Electricity Statistics, Australia (Hydro, Wind, Solar)

Data Provenance

This study uses the monthly electricity statistics of the International Energy Agency (International Energy Agency (IEA), 2022). This data set provides monthly power generation at the national level from 2010 to 2022, measured in gigawatt hours (GWh), covering technologies such as hydropower, wind power, solar power, geothermal power, nuclear power, and fossil fuels (Table 3). For this study, Australian data was selected and we only focused on hydropower, wind power and solar power from July 2010 to December 2022, in order to find out their potential associations with carbon dioxide emissions.

Data Dictionary

Attribute	Data type	Description
Month	Period [M]	Month of measurement
Hydro (GWh)	Float	Hydropower generation
Wind (GWh)	Float	Wind power generation
Solar (GWh)	Float	Solar power generation
Renewables_selected (GWh)	Float	Total monthly generation

Table 3: Data dictionary for Monthly Electricity Statistics, Australia.

Limitations

The dataset reflects only power generation, which may fluctuate due to multiple factors and may not be accurately associated with CO₂ emissions. Therefore, the outcomes could be limited and further study is required to improve the analysis.

2.4 Australian Petroleum Statistics

Data Provenance

This dataset was obtained from the Australian Government Open Data Portal (Australian Government Department of Climate Change, Energy, the Environment and Water, 2025) on 5 September 2025 and provides monthly Australian petroleum statistics covering the period from July 2010 to June 2025. It consists of several worksheets about Australian petroleum production, sales, imports, exports, and prices. For this study, the Sales of products worksheet was chosen since it indirectly reflects the usage of fuel, which could significantly influence the emission of CO₂.

Data Dictionary

The Sales of products worksheet contains 181 rows and 19 columns, and table 4 below presents the description of each attribute.

Attribute	Data type	Description
Month	String	Month of record (MMM YYYY)
Regular (<95 RON) (ML)	Float	Sales volume of regular unleaded petrol with octane rating below 95 RON (in ML)
Premium (95–97 RON) (ML)	Float	Sales volume of premium unleaded petrol with octane rating 95–97 RON (in ML)
Premium (98+ RON) (ML)	Float	Sales volume of premium unleaded petrol with octane rating above 98 RON (in ML)
Ethanol-blended fuel (ML)	Float	Sales volume of ethanol-blended fuel (in ML)
Automotive gasoline total (ML)	Float	Total sales volume of automotive gasoline (in ML)
Diesel oil: total (ML)	Float	Total sales volume of diesel oil (in ML)
Diesel oil: premium diesel (ML)	Float	Sales volume of premium diesel oil (in ML)
Aviation turbine fuel Domestic (ML)	Float	Sales volume of aviation turbine fuel for domestic aviation (in ML)
Aviation turbine fuel International (ML)	Float	Sales volume of aviation turbine fuel for international aviation (in ML)
Aviation turbine fuel total (ML)	Float	Total sales volume of aviation turbine fuel (in ML)
LPG Automotive use (ML)	Float	Sales volume of liquefied petroleum gas (LPG) for automotive use (in ML)
LPG Non-automotive use (ML)	Float	Sales volume of LPG for non-automotive use (in ML)
LPG total (ML)	Float	Total sales volume of LPG (in ML)
Aviation gasoline (ML)	Float	Sales volume of aviation gasoline (in ML)
Fuel oil (ML)	Float	Sales volume of fuel oil (in ML)
Lubricating oils & greases (ML)	Float	Sales volume of lubricating oils and greases (in ML)
Other products (ML)	Float	Sales volume of other petroleum products (in ML)
Total (ML)	Float	Total sales volume of all petroleum products (in ML)

Table 4: Data dictionary for Australian Petroleum Statistics (Sales of Products).

Limitations

The dataset only reports the sales of petroleum products, which cannot directly reflect the actual emissions of CO₂. Moreover, the products sold in a given month may be consumed in subsequent months, leading to potential bias.

3 Data Cleaning

3.1 Cape Grim Greenhouse Gas Data (CO₂)

The dataset was originally in the .csv format, but was imported into Python as .xlsx, and processing was performed using the *pandas* library.

The columns YYYY and MM were merged into a single date column and converted to a monthly period format for subsequent plotting. In addition, all column names were standardized to lowercase to minimize indexing errors.

Outliers were assessed using the rule $\mu \pm 3\sigma$ together with domain knowledge, but none were identified. Only 12 missing values were found in GR (ppm / year) and were handled by mean imputation, while all other columns were complete.

To ensure comparability with collaborator datasets, only observations from July 2010 to December 2022 and the variables date, CO₂ (ppm), and GR (ppm / year) were retained, yielding a cleaned dataset of 150 rows and 3 columns.

3.2 Rainfall and Evaporation

In this dataset, the first step is to read the weather dataset and retain the three core variables of date, rainfall, and evaporation. Delete the variables that are not the focus under this exploration.

Subsequently, the date variable was converted to the standard date format and renamed to facilitate subsequent processing.

For the handling of missing values, missing rainfall values are filled with zero, and abnormal observations less than zero are removed simultaneously. The missing values of evaporation are filled with its overall mean.

After confirming through the statistics of missing values that there are no missing data, we resampled the data by month, summarized the daily data into monthly data, and adjusted the date format to year-month.

Then, to ensure consistency with the data range of other group members, we limited the research period from July 2010 to December 2022.

Finally, to ensure the readability of the result display, we uniformly keep the numerical variables to two decimal places.

3.3 IEA Monthly Electricity Statistics, Australia (Hydro, Wind, Solar)

We first read the IEA monthly electricity dataset and standardized the column headers by trimming whitespace and converting them to lower case to ensure consistency. We then filtered the records to Australia and retained only the three focal products—Hydro, Wind, and Solar—together with the essential fields (year, month, product, value). A monthly time index (Month) was constructed by combining year and month and converting it to a monthly period; the original year and month fields were subsequently dropped to avoid redundancy.

To focus the analysis, ancillary attributes such as `DISPLAY_ORDER`, `CODE_TIME`, `yearToDate`, and `previousYearToDate`, as well as product categories unrelated to the target like coal, oil were removed. The data were then reshaped into a wide monthly table with separate columns for Hydro, Wind, and Solar, and a composite series `Renewables_selected` was computed as their sum.

Finally, to align with the scope of this study and maintain comparability with other team outputs, the sample was restricted to July 2010 through December 2022, and the resulting monthly dataset was exported for subsequent statistics and visualisation.

3.4 Australian Petroleum Statistics

In this section, Python with the *pandas* library was employed to clean the raw dataset, so that the dataset could be convenient and consistent for subsequent merging and analysis.

Firstly, the Month column was converted to a monthly period type in the pandas data frame to be consistent with other datasets and facilitate the date filtering in the subsequent analysis.

Secondly, all irrelevant attributes were removed, retaining only the month and total sales columns. In addition, the column names were converted to lowercase and renamed to ensure consistency and clarity.

Then, appropriate functions were used to examine for missing values, so that the final dataset contained no null entries.

Finally, the data was filtered up to December 2022 to match the datasets provided by other team members.

In the end, the cleaned dataset contained two attributes, month and `petroleum_total_ml`, with a total of 150 records.

4 Simple Analysis

4.1 Cape Grim Greenhouse Gas Data (CO₂)

Between 2010 and 2022, the mean CO₂ concentration was approximately 400.72 ppm, rising from 386.57 ppm in 2010 to 415.34 ppm in 2022. The standard deviation of 8.70 ppm reflects a steady and consistent upward trend (Table 5).

Index	Count	Max	Min	Mean	Median	Std. Dev.
CO ₂ (ppm)	150	415.34	386.57	400.72	401.30	8.70
GR (ppm/yr)	150	3.54	1.38	2.32	2.28	0.45

Table 5: Descriptive statistics of CO₂ concentration and growth rate (2010–2022).

Correspondingly, the monthly mean CO₂ trend exhibits a sawtooth pattern with a steady rise (Figure 1), driven primarily by seasonal processes such as the uptake and release of CO₂ by terrestrial vegetation. Nonetheless, the long-term trajectory remains one of sustained growth.

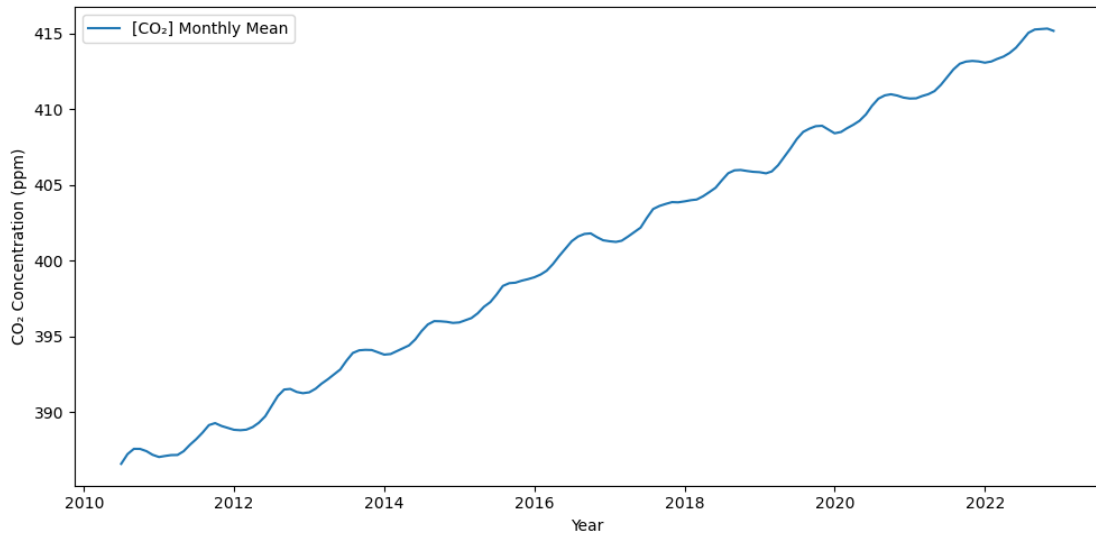


Figure 1: Trend of Monthly Mean CO₂ Concentrations (2010–2022).

During the same period, the annual CO₂ growth rate ranged from 1.38 to 3.54 ppm / year, with a mean of 2.32 ppm / year and a standard deviation of 0.45 ppm / year, reflecting pronounced interannual variability (Table 5).

At the decadal scale, the mean growth rate also exhibited an upward trajectory (Figure 2). This indicates that CO₂ concentrations are not only increasing, but the rate of increase is itself accelerating, a pattern that can be preliminarily attributed to anthropogenic activities.

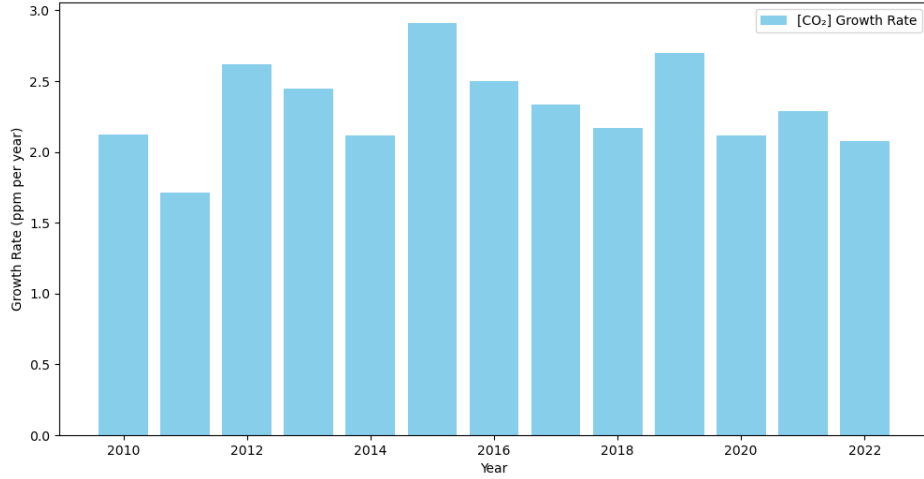


Figure 2: Annual Mean Growth Rate of CO₂ (2010–2022).

4.2 Rainfall and Evaporation

Index	Count	Max	Min	Mean	Median	Std. Dev.
Rainfall	150	332.4	4.4	84.68	74.15	59.32
Evapotranspiration	150	171.1	17.0	72.23	63.80	46.98

Table 6: Descriptive statistics of rainfall and evapotranspiration (2010–2022).

Table 6 presents statistical indicators of rainfall and evaporation, showing that the distribution of rainfall is highly uneven, ranging from 4.4mm to 332.4mm, with a large standard deviation (59.32), indicating significant fluctuations in monthly rainfall and its susceptibility to seasonal cycles and extreme weather events. In contrast, the variation in evapotranspiration is relatively small, with a standard deviation of 46.98. The overall value is more stable, reflecting that it is mainly driven by long-term climatic conditions and solar radiation and other factors, and does not change as drastically as rainfall. Overall, the regional water balance is mainly determined by the uncertainty of rainfall, which means that sudden fluctuations in rainfall may have a significant impact on soil moisture content and agricultural production. Meanwhile, changes in rainfall may also further amplify the fluctuations of CO₂ by affecting vegetation growth and soil respiration.

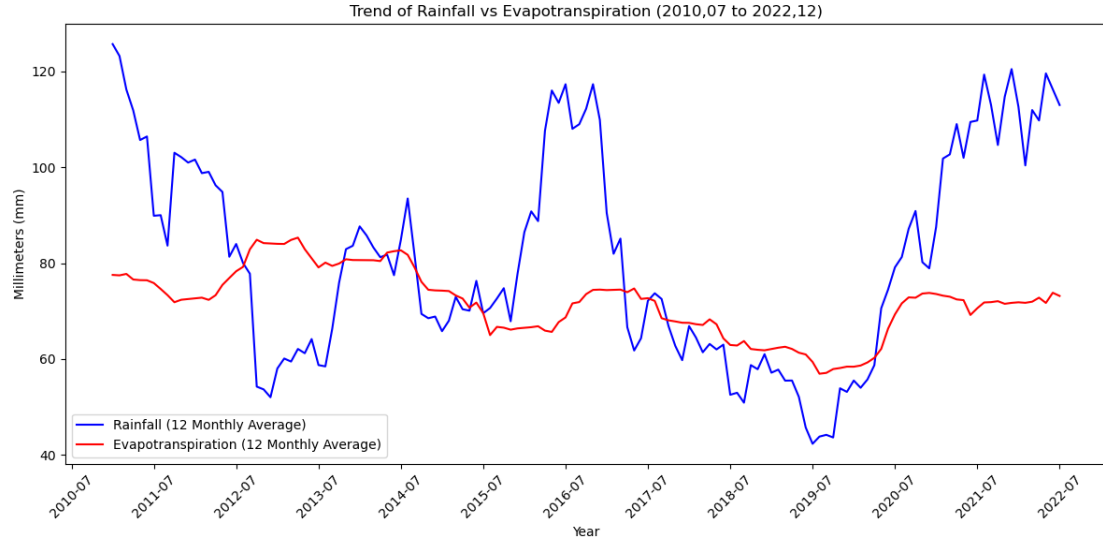


Figure 3: Smoothed monthly rainfall and evapotranspiration trends (2010–2022).

For the rainfall and evapotranspiration (Figure 3) Rainfall shows strong peaks in 2012, 2016, 2020 and 2022 (climate extremes), while evapotranspiration stays steady. This imbalance highlights rainfall as a key uncertainty factor.

4.3 IEA Monthly Electricity Statistics, Australia (Hydro, Wind, Solar)

Index	Count	Max	Min	Mean	Median	Std. Dev.
Production	150	8979.50	1514.95	3720.06	3130.17	1737.10

Table 7: Descriptive statistics of Australia Hydro+Wind+Solar (monthly, 2010-07 to 2022-12).

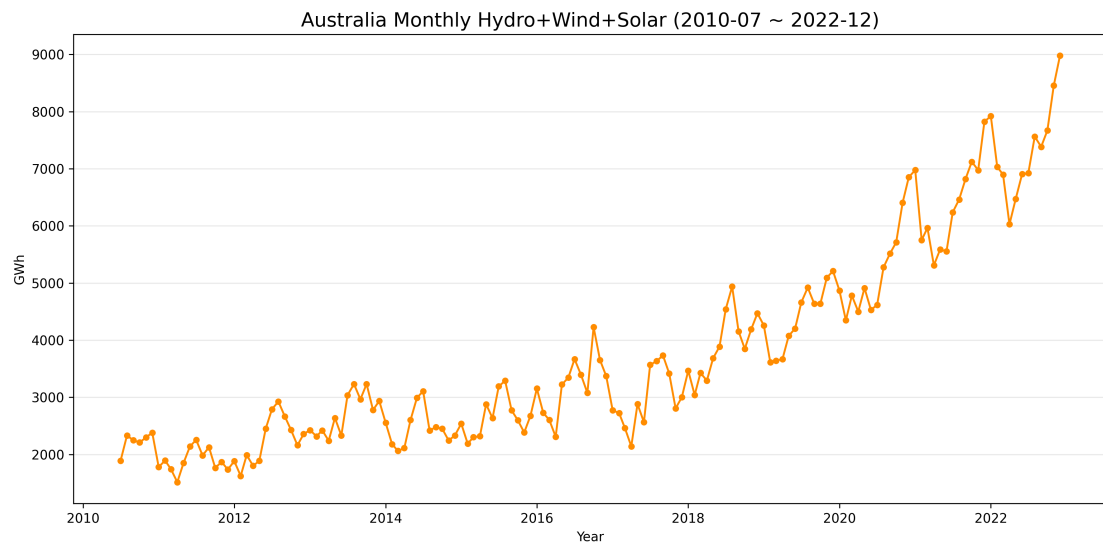


Figure 4: Monthly Hydro+Wind+Solar (GWh), Australia, Jul 2010 – Dec 2022.

Figure 4 shows the monthly changes in hydropower, wind power and solar power generation in Australia. However, from mid-2010 to the end of 2022, the fundamental trend rose steadily. Profits in the early 2010s were not significant. With the increase in new production capacity, it became evident around 2017-2019 and accelerated in 2020 and beyond. Seasonal weather patterns cause brief drops, but usually rise to a higher altitude each year than the previous one. By the end of 2022, monthly power generation will reach approximately 9 terawatt hours, several times that of 2010. This indicates that the structural expansion of renewable energy is continuous rather than a temporary fluctuation.

4.4 Australian Petroleum Statistics

Index	Count	Max	Min	Mean	Median	Std. Dev.
Total sales	150	5361.4	3399.2	4745.6	4771.6	300.9

Table 8: Descriptive Statistics of the Monthly Petroleum Sales.

Table 8 summarizes the descriptive statistics of the cleaned petroleum sales dataset. The dataset contains 150 monthly observations and it spans a range from 3399.2 ML to 5361.4 ML, presenting considerable fluctuations. The mean value is 4745.6 ML, which is close to the median of 4771.6 ML, indicating a relatively symmetric distribution. The standard deviation of 300.9 shows noticeable variability in monthly petroleum sales.

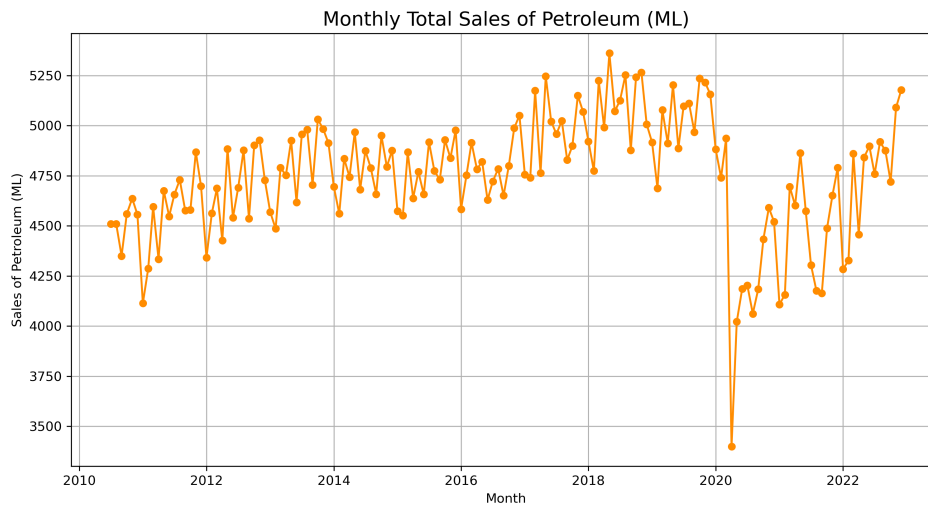


Figure 5: Monthly Total Sales of Petroleum (ML) from July 2010 to December 2022.

Matplotlib library was used to draw line graph in this part. Figure 5 demonstrates the variability of monthly total sales of petroleum. Overall, the sales presented an upward trend over the 12 years. Before 2020, although notable fluctuations were observed between each month, sales still gradually rose. However, a significant decline happened in April 2020, when

the minimum value was witnessed. The decrease was highly correlated with the COVID-19 pandemic, which resulted in a decrease in the demand for transportation and therefore led to lower consumption of petroleum. After that, sales showed an increasing trend again with fluctuations and slowly recovered to normal level.

4.5 Combined Summary & Comparison

After integrating four datasets into a unified time series (2010–2022), the dataset included atmospheric CO₂ concentration (ppm), its annual growth rate (ppm/year), hydro, solar, wind, aggregated renewables generation, petroleum sales, and hydrological indicators.

Correlation analysis initially revealed strong associations of CO₂ with wind ($r = 0.9245$), solar ($r = 0.8428$), and total renewables ($r = 0.8953$), whereas its correlation with hydro was negligible ($r = -0.0099$). Petroleum sales showed only weak associations with CO₂ concentration ($r = 0.06$) and its growth rate ($r = 0.16$), and negligible or negative associations with renewables. When first differences were applied to remove the common time trend, these correlations declined substantially, with ΔCO_2 vs $\Delta\text{Renewables}$ at $r = 0.2703$ and ΔCO_2 vs ΔSolar nearly null ($r = -0.0001$). Year-on-year growth rates yielded even weaker relationships, with the strongest between CO₂ growth and hydro growth ($r = 0.3849$), while solar growth was negatively correlated ($r = -0.1697$). Hydrological indicators such as rainfall and evaporation showed little association with CO₂, further underscoring that non-trend variables contributed minimally.

Finally, a multiple regression model incorporating both the trend in time and total renewables explained almost all variations in CO₂ ($R^2 = 0.9967$). In this model, the time trend coefficient was 0.195823, while the renewables coefficient was minimal (0.000116), indicating that the upward trajectory of CO₂ was driven primarily by the long-term temporal trend rather than renewables generation. After controlling for the time trend, the partial correlation between CO₂ and renewables was reduced to 0.3743, suggesting much of the apparent association in the raw data reflected a shared trend over time rather than a direct causal relationship.

5 Conclusion

5.1 Summary

Our analysis demonstrates that the rise in Australia’s CO₂ emissions over the past decade has been driven mainly by long-term structural dynamics rather than short-term shifts in renewable deployment, petroleum use, or climate variability. Although renewables and emissions appear strongly correlated, this link is largely an artefact of their parallel upward paths over time, not direct causation. Thus, mere renewable expansion alone is unlikely to markedly alter Australia’s emissions path. For policymakers and industry, it is vital to recognize that even under accelerated renewable uptake, meeting the 2030 and 2050 targets will depend chiefly on policies that transform fossil fuel infrastructure, moderate demand, and cut industrial emissions, not on sole reliance on supply-side renewable growth (CSIRO & ClimateWorks Australia, 2023).

5.2 Contribution List

Define Topic Text Part: Yuhang Zhang, Siqi Wang

Cape Grim Greenhouse Gas Data (CO₂) Simple Analysis & Text Part: Siqi Wang

Rainfall and Evaporation Simple Analysis & Text Part: Yuhang Zhang

IEA Monthly Electricity Statistics, Australia (Hydro, Wind, Solar) Simple Analysis & Text Part: Kyle Huang

Australian Petroleum Statistics Simple Analysis & Text Part: Jaron Shen

Data Integration: Jaron Shen

Correlation Analysis: Siqi Wang

Combined Summary & Comparison Text Part: Siqi Wang

Conclusion Text Part: Siqi Wang

Reference List: Kyle Huang

Report Standardization: Jaron Shen

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